

THE BEST DISSERTATION YOU WILL EVER READ
APACHE POINT OBSERVATORY FOLLOW-UP OF
ACCELERATING CANDIDATE EXOPLANET HOST STARS (APO
ACCELS)
AND CHARACTERIZATION OF THE BINARY HOST OF THE DIRECTLY
IMAGED PLANET HD 143811 AB b

by

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A dissertation submitted to the Graduate School

in partial fulfillment of the requirements

for the degree

Doctor of Philosophy

Major Subject: Astronomy

New Mexico State University

Las Cruces, New Mexico

August 2026

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ACKNOWLEDGMENTS

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PUBLICATIONS

Peck, A., Roberson, W., Nielsen, E., De Rosa, R., Jones, N., Wang, J., Macintosh, B., Lewis, B., Duchêne, G., Metchev, S., Abbas, A., Xuan, J., Sanghi, A., Patience, J., Barman, T., Bulger, J., Chilcote, J., Esposito, T., Fitzgerald, M., Follette, K., Gallamore, H., Goodsell, S., Graham, J., Greenbaum, A., Hibon, P., Ingraham, P., Kalas, P., Konopacky, Q., Marchis, F., Maire, J., Marois, C., Matthews, B., Mawet, D., Millar-Blanchaer, M., Oppenheimer, R., Palmer, D., Perrin, M., Poyneer, L., Pueyo, L., Rajan, A., Rameau, J., Rantakyö, F., Ren, B., Ruffio, J.B., Savransky, D., Schneider, A., Sivaramakrishnan, A., Smith, A., Song, I., Soummer, R., Thomas, A., Ward-Duong, K., Wolff, S., *Characterization of the Host Binary of the Directly Imaged Exoplanet HD 143811 AB b*. ApJ Letters, 995, 40, doi: 10.3847/2041-8213/ae1a67

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CONFERENCE PROCEEDINGS

AWARDS AND GRANTS

2021–2023	NMSU Astronomy William Webber Voyager Graduate Fellowship
2025–2026	NMSU Astronomy Pegasus Award for excellence in teaching
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LEADERSHIP AND PROFESSIONAL EXPERIENCE

ABSTRACT

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Dr. Eric Nielsen, Advisor

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LIST OF ABBREVIATIONS

Instruments, Telescopes & Observatories

Acronyms and Jargon

LIST OF SYMBOLS

Symbol	Meaning
$q = M_2/M_1, \ M_1 > M_2$	mass ratio
M_1	mass of more massive binary component
M_2	mass of less massive binary component

1. TITLE

2. Combined Age Posteriors

For stars with $B - V \leq 0.58$ (corresponding to an F9.5 spectral type and approximate main-sequence mass of $1.1M_{\odot}$), we apply additional constraints on their ages based on stellar lifetimes and their CMD positions (e.g. ??). Using this technique, we also produce age posteriors for stars too massive for Li, R'_{HK} , or gyrochronology. CMD age posteriors are especially useful in cases where we can only derive a lower limit on the age from a lithium equivalent width upper limit because the stellar lifetime can be used to constrain the upper bound on the age (e.g. late F stars). To calculate CMD position-based age posteriors, we follow the method from ?, which uses a Bayesian approach to compute ages from stellar models and optical photometry. For each star, MCMC is used to explore the 5-dimensional parameter space in stellar mass, age, metallicity, initial rotation, inclination of the rotation axis, and parallax. We use the same priors on these parameters as in ?, and a Gaussian likelihood for the *Gaia* G_B , G , and G_R catalog photometry and errors. The model predictions at each MCMC step are generated, as in ?, from a combination of MESA (?) evolutionary models with ATLAS9 model atmospheres (?). As in ?, for stars with $B - V \leq 0.27$, we include a prior based on the metallicity distribution of stars < 1 Gyr from ?.

For 55 stars more than one age tracer is available, and we improve our con-

straints by taking the product of each age posterior for a given star. In Table 2.1 we present our full 166-star sample, including basic properties and final median ages and 68% and 95% confidence intervals for each star. Figure 2.1 shows the age posteriors for the sample as a function of $B - V$.

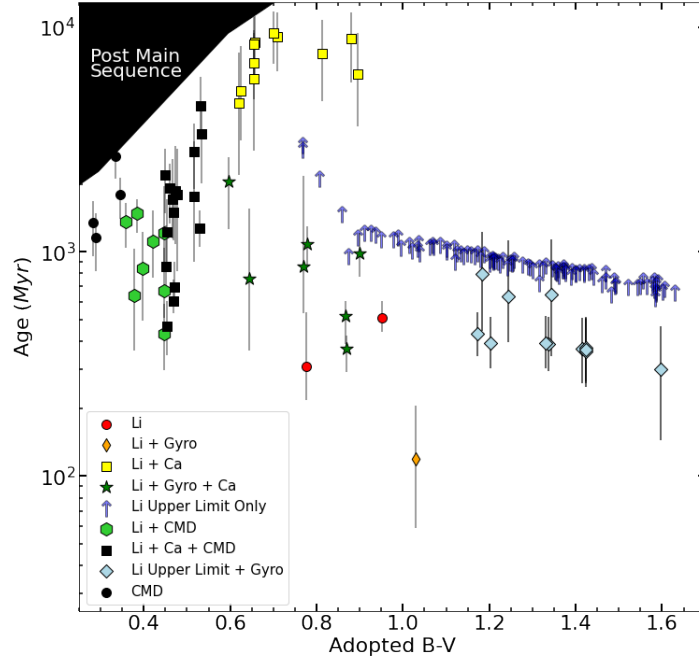


Fig. 2.1.— The medians and 68% confidence intervals for our derived age posteriors for the full sample of 166 stars, as a function of $B - V$. The symbol and color of each star represent which age tracer (or tracers) was used to derive the age posterior. The error bars represent the 68% confidence intervals.

Table 2.1:: Our complete sample of accelerating stars with age posteriors and age tracers used. Ages come from lithium (Li), lithium upper limits (Li UL), calcium emission (R’HK), rotation periods (Gyro), and CMD positions (CMD).

Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 143	00 01 49.35	+21 27 45.01	9.9811 ± 0.0002	1.033 ± 0.019	K3	43.4 ± 0.59	Li UL	≥ 4442.08	≥ 1034.76	≥ 484.61
HIP 669	00 08 16.36	-14 49 28.16	6.8956 ± 0.0002	0.624 ± 0.015	G1V	27.31 ± 0.02	Li, R’HK	5234.81	3132.71– 8290.74	1783.08– 11668.88
HIP 1068	00 13 15.83	+69 19 36.76	12.0476 0.0015	$\pm 1.315 \pm 0.1$	M4	20.52 ± 0.05	Li UL	≥ 4217.04	≥ 720.14	≥ 138.08
HIP 1412	00 17 40.90	-08 40 56.14	10.2711 0.0006	$\pm 1.414 \pm 0.005$	K7V	32.05 ± 0.09	Li UL, Gyro	367.99	258.82–508.19	148.22–612.04
HIP 1539	00 19 12.39	-03 03 13.01	10.2808 0.0004	$\pm 1.375 \pm 0.005$	K5V	31.19 ± 0.02	Li UL	≥ 4294.84	≥ 828.91	≥ 280.46
HIP 1771	00 22 25.19	+06 40 03.16	10.6176 0.0005	$\pm 1.489 \pm 0.1$	K7V	43.88 ± 0.09	Li UL	≥ 4234.75	≥ 744.83	≥ 164
HIP 2426	00 30 55.58	+73 22 14.89	10.0247 0.0002	$\pm 1.359 \pm 0.02$	K	29.48 ± 0.01	Li UL	≥ 4298.94	≥ 834.64	≥ 285.77
HIP 2843	00 36 01.85	-05 34 14.58	6.6114 ± 0.0002	0.466 ± 0.007	F6V	44.84 ± 0.06	Li UL, R’HK, CMD	1715.26	1150.23– 2585.26	808.98–3587.63
HIP 3008	00 38 15.29	+52 19 55.52	9.8115 ± 0.0003	1.424 ± 0.02	M0.0V	23.02 ± 0.01	Li UL, Gyro	360.24	250.15–502.94	128.65–609.82
HIP 3293	00 42 01.42	+40 41 17.71	6.9164 ± 0.0003	0.452 ± 0.009	F5	49.58 ± 0.07	Li, R’HK, CMD	862.22	651.1–1311.35	521.75–2077.47
HIP 3633	00 46 34.04	+14 19 31.61	9.6005 ± 0.0001	0.814 ± 0.061	K3V	46.51 ± 0.21	Li UL, R’HK	7664.88	4696.93– 10795.26	2383.87– 12584.18
Continued on next page										

Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 11815	02 32 23.18	+03 22 57.48	11.2101 0.0014	$\pm 1.357 \pm 0.02$	K5	49.26 ± 0.05	Li UL	≥ 4276.65	≥ 803.47	≥ 253.41
HIP 12837	02 45 01.09	-22 09 59.51	7.0178 ± 0.0002	0.517 ± 0.008	F7V	38.9 ± 0.03	Li, R'HK, 1771.7 CMD		899.96–3111.02	386.64–4669.22
HIP 13681	02 56 14.10	-11 50 49.84	9.8618 ± 0.0003	1.062 ± 0.015	—	47.51 ± 0.04	Li UL	≥ 4456.4	≥ 1054.8	≥ 509.53
HIP 13782	02 57 23.76	-23 51 43.79	5.399 ± 0.0005	0.238 ± 0.003	A5IV/V	48.69 ± 0.15	CMD	757.31	595.43–917.22	443.02–1075.26
HIP 15211	03 16 05.89	-05 51 15.85	9.9066 ± 0.0002	0.936 ± 0.007	K3	49.29 ± 0.04	Li UL	≥ 4551.82	≥ 1187.49	≥ 609.82
HIP 16247	03 29 22.87	-24 06 03.09	8.8107 ± 0.0005	1.02 ± 0.037	K3VkFe+0.4	31.08 ± 0.02	Li UL	≥ 4428.57	≥ 1015.84	≥ 461.81
HIP 16709	03 34 59.64	+34 36 39.51	10.3002 0.0004	$\pm 1.225 \pm 0.02$	M0	41.1 ± 0.65	Li UL	≥ 4331.28	≥ 879.86	≥ 328.75
HIP 17118	03 39 58.81	+63 52 13.47	6.7675 ± 0.0002	0.478 ± 0.023	F5	42.81 ± 0.03	Li, R'HK, 1790.5 CMD		818.76–2885.78	405.89–3903.25
HIP 17213	03 41 13.71	+75 23 56.51	9.6211 ± 0.0006	1.1 ± 0.015	K0	43.63 ± 0.08	Li UL	≥ 4420.43	≥ 1004.51	≥ 453.14
HIP 18527	03 57 43.92	-20 16 04.05	8.6088 ± 0.0004	0.899 ± 0.021	K2VkFe–0.5	37.38 ± 0.03	Li UL, Gyro, 981.67 R'HK		770.11–1216.88	619.34–1469.49
HIP 19739	04 13 55.87	+82 55 06.98	10.1025 0.0005	$\pm 1.424 \pm 0.02$	M0.0V	36.94 ± 0.17	Li UL, Gyro	372.38	265.37–510.32	199.63–612.75
HIP 19912	04 16 18.56	+08 43 09.96	9.4283 ± 0.0002	1.148 ± 0.002	K7V	38.96 ± 0.26	Li UL	≥ 4408.38	≥ 987.64	≥ 433.87
HIP 20419	04 22 25.69	+11 18 20.56	9.3585 ± 0.0004	1.183 ± 0.005	K2	44.05 ± 0.48	Li UL, Gyro	797.7	444.12–1230.22	313.95–1599.99
HIP 20648	04 25 29.38	+17 55 40.46	4.2957 ± 0.001	0.049 ± 0.007	A2IV–Vs	47.42 ± 0.31	CMD	602.61	530.07–670.92	456.89–742.38
HIP 21066	04 30 57.17	+10 45 06.36	6.9211 ± 0.0002	0.472 ± 0.013	F7V	46.51 ± 0.06	Li UL, R'HK, 1864.19 CMD		1085.7–2959.28	650.55–3268.37
HIP 21152	04 32 04.80	+05 24 36.14	6.2678 ± 0.0002	0.42 ± 0.014	F5V	43.27 ± 0.05	Li UL, CMD	1116.14	765.3–1521.58	423.86–1861.12

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Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 40518	08 16 23.06	-17 22 03.83	10.0726 0.0003	$\pm 1.134 \pm 0.004$	—	46.43 ± 0.6	Li UL	≥ 4406.56	≥ 985.1	≥ 430.98
HIP 41274	08 25 14.08	-07 10 12.85	7.0892 ± 0.0006	0.952 ± 0.011	K0V	49.14 ± 0.07	Li	505.8	439.44–602.43	335.92–820.22
HIP 41277	08 25 16.90	+04 15 12.72	9.9158 ± 0.002	1.03 ± 0.015	K8V	41.68 ± 0.08	Li, Gyro	119.62	58.63–206.48	17.47–300.01
HIP 41319	08 25 49.87	+17 02 46.57	5.9986 ± 0.0003	0.448 ± 0.005	F5IIIm?	45.5 ± 0.07	Li, CMD	668.73	509.36–1040.76	400.75–1505.21
HIP 42231	08 36 37.96	+39 02 13.55	11.2749 0.0003	$\pm 1.241 \pm 0.015$	—	43.88 ± 0.15	Li UL	≥ 4308.4	≥ 847.87	≥ 296.87
HIP 42343	08 37 57.55	-08 52 53.72	6.9078 ± 0.0002	0.516 ± 0.001	F6V	48.73 ± 0.05	Li, R'HK, CMD	2791.29	1896.43–3724.4	1184.85–4152.26
HIP 42507	08 40 00.26	-06 28 33.06	9.2424 ± 0.0003	1.361 ± 0.001	K6V	25.46 ± 0.09	Li UL	≥ 4259.1	≥ 778.95	≥ 226.84
HIP 43745	08 54 35.72	+63 45 29.74	10.4509 0.0002	$\pm 1.423 \pm 0.02$	K5V	34.26 ± 0.02	Li UL, Gyro	366.55	260.53–505.31	168.77–610.43
HIP 44162	08 59 39.91	-19 12 28.95	6.0845 ± 0.0003	0.461 ± 0.005	F5V	40.24 ± 0.13	Li UL, R'HK, CMD	1925.01	1340.03–2478.25	901.96–3477.51
HIP 44953	09 09 30.16	-00 23 54.35	8.4254 ± 0.0002	0.767 ± 0.002	G8V	34.95 ± 0.03	Li UL	≥ 5993.75	≥ 2846.19	≥ 1323.58
HIP 45621	09 17 55.38	-03 23 14.09	7.5442 ± 0.0003	0.869 ± 0.013	K1/2V	32.28 ± 0.04	Li, Gyro, R'HK	371.47	291.48–423.05	144.29–489.92
HIP 45731	09 19 22.86	+62 03 16.89	10.5142 0.0006	$\pm 1.586 \pm 0.02$	M1.0Ve	37.94 ± 0.03	Li UL	≥ 4167.6	≥ 650.87	≥ 91.63
HIP 46385	09 27 30.56	+50 39 12.45	10.9763 0.0005	$\pm 1.586 \pm 0.1$	M2.5V	31.16 ± 0.05	Li UL	≥ 4175.04	≥ 661.4	≥ 92.44
HIP 47013	09 34 53.52	+72 12 20.44	5.6501 ± 0.0005	0.53 ± 0.003	F7V	49.6 ± 0.08	Li UL, R'HK, CMD	1275.94	1054.99–1523.22	861.32–1758.82
HIP 47261	09 37 58.33	+22 31 23.15	9.4588 ± 0.0003	1.21 ± 0.015	K4V	30.89 ± 0.02	Li UL	≥ 4351.49	≥ 908.11	≥ 356.79

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Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 47300	09 38 21.77	+40 14 23.25	5.237 ± 0.0007	0.223 ± 0.004	F0V	38.49 ± 0.12	CMD	718.62	500.33–934.73	298.93–1134.22
HIP 47387	09 39 14.59	+66 50 58.14	9.6883 ± 0.0002	1.07 ± 0.047	K2	41.63 ± 0.54	Li UL	≥ 4450.55	≥ 1046.57	≥ 489.6
HIP 47539	09 41 31.44	-06 23 35.83	11.2695 ± 0.0004	$\pm 1.353 \pm 0.011$	—	44.42 ± 0.37	Li UL	≥ 4288.12	≥ 819.51	≥ 270.07
HIP 47741	09 43 55.61	+26 58 08.40	10.8486 ± 0.0004	$\pm 1.542 \pm 0.039$	M3.5V	14.37 ± 0.01	Li UL	≥ 4202.01	≥ 698.97	≥ 130.85
HIP 48165	09 49 03.16	+07 49 08.68	9.5048 ± 0.0002	0.925 ± 0.015	G5	45.77 ± 0.48	Li UL	≥ 4574.64	≥ 1219.3	≥ 638.59
HIP 48629	09 54 59.84	+40 23 07.28	10.7594 ± 0.0002	$\pm 1.581 \pm 0.197$	K5V	37.37 ± 0.04	Li UL	≥ 4234.47	≥ 744.26	≥ 152.31
HIP 49046	10 00 26.71	+27 16 01.61	10.5279 ± 0.0002	$\pm 1.372 \pm 0.015$	M0.5V	36.01 ± 0.02	Li UL	≥ 4292.26	≥ 825.31	≥ 276.52
HIP 49526	10 06 43.80	+41 42 52.65	10.5229 ± 0.0003	$\pm 1.492 \pm 0.02$	M1.0V	22.28 ± 0.02	Li UL	≥ 4210.41	≥ 710.86	≥ 147.69
HIP 49577	10 07 13.74	-14 18 16.54	9.6168 ± 0.0004	1.331 ± 0.014	K5V	39.53 ± 0.64	Li UL	≥ 4322.44	≥ 867.5	≥ 318.63
HIP 49910	10 11 23.70	-01 39 44.69	11.1457 ± 0.0004	$\pm 1.391 \pm 0.023$	—	46.55 ± 0.04	Li UL	≥ 4286.95	≥ 817.89	≥ 268.72
HIP 51208	10 27 34.85	+12 58 48.14	10.2039 ± 0.0003	$\pm 1.2 \pm 0.015$	K4V	46.38 ± 0.46	Li UL	≥ 4361.89	≥ 922.66	≥ 371.01
HIP 52339	10 41 48.44	+12 06 31.90	10.9365 ± 0.0003	$\pm 1.42 \pm 0.015$	K7V	37.04 ± 0.03	Li UL	≥ 4281.04	≥ 809.62	≥ 260.19
HIP 53175	10 52 39.61	+00 29 01.95	10.1771 ± 0.0004	$\pm 1.293 \pm 0.005$	K4V	39.35 ± 0.03	Li UL	≥ 4312.99	≥ 854.28	≥ 304.58
HIP 53869	11 01 16.35	+29 35 27.47	9.4421 ± 0.0003	0.868 ± 0.045	K3V	49.58 ± 0.06	Li UL, Gyro, 515.87 R'HK		388.61–604.35	245.7–648.74

Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 59953	12 17 48.39	+46 37 20.73	10.7356 0.0002	$\pm 1.602 \pm 0.1$	M0.5V	35.68 ± 0.02	Li UL	≥ 4177.73	≥ 665.13	≥ 96.84
HIP 60829	12 28 04.06	-15 39 07.60	8.1604 ± 0.0002	0.71 ± 0.012	G6V	47.06 ± 0.06	Li UL, R'HK	9106.19	6391.7– 11635.89	4156.82– 12789.88
HIP 61947	12 41 46.81	+43 02 26.34	11.1443 0.0003	$\pm 1.582 \pm 0.082$	K5V	44.66 ± 0.04	Li UL	≥ 4183.93	≥ 673.77	≥ 104.87
HIP 62325	12 46 22.54	+09 32 22.86	5.3697 ± 0.0004	0.989 ± 0.007	G9	43.71 ± 0.12	Li UL	≥ 4529.3	≥ 1156.33	≥ 585.8
HIP 62627	12 49 56.23	+71 11 39.30	9.707 ± 0.0001	0.995 ± 0.003	K8	49.26 ± 0.07	Li UL	≥ 4447.71	≥ 1042.53	≥ 480.59
HIP 63419	12 59 45.50	-04 25 49.04	8.4878 ± 0.0004	0.779 ± 0.021	K1V	37.11 ± 0.04	Li, Gyro, R'HK	1081.58	874.63–1294.12	662.26–1603.98
HIP 63421	12 59 47.34	-06 51 37.39	11.1775 0.0005	$\pm 1.354 \pm 0.01$	K5V	43.46 ± 0.04	Li UL	≥ 4285.0	≥ 815.16	≥ 265.57
HIP 63661	13 02 51.36	-02 40 26.25	10.9351 0.0003	$\pm 1.468 \pm 0.1$	K5V	48.05 ± 0.06	Li UL	≥ 4236.36	≥ 747.11	≥ 166.58
HIP 65120	13 20 43.83	+10 52 33.31	10.8352 0.0003	$\pm 1.49 \pm 0.02$	M0V	31.35 ± 0.03	Li UL	≥ 4206.97	≥ 706.06	≥ 142.13
HIP 65602	13 27 02.90	-24 17 25.56	8.4679 ± 0.0002	0.911 ± 0.002	K2+V	29.87 ± 0.03	Li UL	≥ 4574.65	≥ 1219.41	≥ 641.92
HIP 65651	13 27 35.55	-01 50 16.98	10.3186 0.0002	$\pm 1.203 \pm 0.02$	K4	45.52 ± 0.05	Li UL, Gyro	392.63	301.17–513.07	126.99–611.37
HIP 65706	13 28 17.76	+30 02 45.98	10.5091 0.0003	$\pm 1.577 \pm 0.082$	K5V	40.49 ± 0.09	Li UL	≥ 4215.51	≥ 717.64	≥ 144.27
HIP 66262	13 34 49.79	+47 22 34.35	9.8141 ± 0.0001	1.165 ± 0.067	K4V	36.82 ± 0.36	Li UL	≥ 4386.44	≥ 956.97	≥ 393.3
HIP 66315	13 35 29.15	+46 33 30.88	10.4651 0.0002	$\pm 1.039 \pm 0.121$	K3	43.23 ± 0.03	Li UL	≥ 4412.74	≥ 985.51	≥ 395.09

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Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 66587	13 38 58.68	-06 14 12.46	10.0215 0.0003	$\pm 1.419 \pm 0.009$	M0.5V	24.19 ± 0.01	Li UL	≥ 4273.57	≥ 799.18	≥ 249.51
HIP 66828	13 41 47.23	+21 29 08.18	10.3186 0.0002	$\pm 1.33 \pm 0.02$	K5V	38.3 ± 0.03	Li UL, Gyro	393.32	301.47–515.92	223.58–613.17
HIP 69311	14 11 12.64	+43 06 23.96	9.5996 ± 0.0002	1.172 ± 0.07	K5V	35.34 ± 0.01	Li UL, Gyro	431.78	342.15–536.81	251.18–620.33
HIP 69333	14 11 24.19	+30 05 02.44	9.8627 ± 0.0002	1.173 ± 0.004	K5V	43.8 ± 0.03	Li UL	≥ 4402.4	≥ 979.3	≥ 426.36
HIP 69860	14 17 47.84	+21 26 01.58	10.8113 0.0005	$\pm 1.629 \pm 0.1$	M1.0V	41.84 ± 0.21	Li UL	≥ 4171.9	≥ 656.97	≥ 92.18
HIP 69963	14 18 59.04	+38 38 26.33	10.7754 0.0004	$\pm 1.39 \pm 0.02$	M1.0V	31.69 ± 0.05	Li UL	≥ 4252.64	≥ 769.92	≥ 217.68
HIP 70472	14 24 49.86	-17 27 08.09	10.1555 0.0004	$\pm 1.257 \pm 0.01$	K7V	34.2 ± 0.05	Li UL	≥ 4358.91	≥ 918.5	≥ 368.06
HIP 71243	14 34 11.70	+32 32 04.12	6.2142 ± 0.0001	0.45 ± 0.003	F5V	41.36 ± 0.03	Li UL, R'HK, CMD	2202.78	1488.73–2873.15	911.32–3350.92
HIP 71515	14 37 33.07	-16 32 27.62	8.4606 ± 0.0005	0.95 ± 0.002	K2/3V	44.11 ± 0.3	Li UL	≥ 4516.62	≥ 1138.55	≥ 566.76
HIP 71899	14 42 23.10	+21 17 35.10	7.3501 ± 0.0002	0.533 ± 0.01	F8	45.58 ± 0.04	Li, R'HK, CMD	3344.07	2011.77–4550.69	935.24–5494.66
HIP 71957	14 43 03.62	-05 39 29.53	3.7671 ± 0.002	0.385 ± 0.006	F2V	19.16 ± 0.14	Li, CMD	1483.42	1211.67–1716.6	893.61–1957.38
HIP 73121	14 56 40.64	+21 04 16.72	9.1024 ± 0.0002	1.035 ± 0.009	K	40.88 ± 0.16	Li UL	≥ 4483.97	≥ 1093.34	≥ 544.57
HIP 73183	14 57 25.90	+55 54 39.01	9.9311 ± 0.0006	1.555 ± 0.197	K5V	37.04 ± 0.4	Li UL	≥ 4236.44	≥ 747.08	≥ 152.29
HIP 73252	14 58 15.67	+59 35 00.43	9.5255 ± 0.0002	1.565 ± 0.082	K5V	25.75 ± 0.01	Li UL	≥ 4205.13	≥ 703.31	≥ 129.34
HIP 73787	15 04 53.98	-18 35 27.36	9.006 ± 0.0004	1.21 ± 0.008	K5+Vk	33.0 ± 0.64	Li UL	≥ 4358.02	≥ 917.25	≥ 366.04
HIP 74926	15 18 39.55	-18 37 35.71	9.8387 ± 0.0003	1.214 ± 0.009	K4V	27.35 ± 0.1	Li UL	≥ 4350.77	≥ 907.11	≥ 356.02

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Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 79203	16 09 55.27	-18 20 26.25	6.3266 ± 0.0003	0.447 ± 0.007	F3V	49.7 ± 0.17	Li, CMD	430.94	297.69–584.21	170.71–1095.7
HIP 79593	16 14 20.73	-03 41 39.56	2.0164 ± 0.0055	1.584 ± 0.01	M0.5III	49.0 ± 1.33	Li UL	≥ 4144.8	≥ 619.13	≥ 62.8
HIP 81655	16 40 48.88	+36 19 00.08	10.5661 ± 0.0003	$\pm 1.605 \pm 0.1$	M2V	21.47 ± 0.02	Li UL	≥ 4191.94	≥ 684.91	≥ 114.09
HIP 82260	16 48 28.31	-16 20 04.33	7.4515 ± 0.0002	0.767 ± 0.002	G8IV–V	37.01 ± 0.18	Li UL	≥ 6115.02	≥ 2980.3	≥ 1383.07
HIP 83676	17 06 08.20	-06 10 02.07	8.489 ± 0.0003	1.015 ± 0.015	K3V	30.22 ± 0.02	Li UL	≥ 4468.19	≥ 1071.18	≥ 513.21
HIP 83929	17 09 19.61	+41 26 09.85	10.4487 ± 0.0004	$\pm 1.597 \pm 0.197$	K4V	39.68 ± 0.06	Li UL, Gyro	298.37	145.14–467.64	84.98–595.32
HIP 87370	17 51 07.73	-22 55 15.61	6.991 ± 0.0001	0.656 ± 0.015	G3V	44.27 ± 0.19	Li, R'HK	8392.46	5631.85–11082.4	3331.8–12696.46
HIP 88962	18 09 33.26	-12 02 19.98	9.8511 ± 0.0005	1.372 ± 0.004	K7V	27.27 ± 0.01	Li UL	≥ 4310.61	≥ 850.97	≥ 302.95
HIP 89320	18 13 30.98	+81 04 32.78	10.5168 ± 0.0003	$\pm 1.586 \pm 0.197$	K8	43.84 ± 0.02	Li UL	≥ 4209.43	≥ 709.43	≥ 119.63
HIP 89449	18 15 18.29	+18 29 59.98	9.5074 ± 0.0003	1.323 ± 0.019	K7V	29.19 ± 0.01	Li UL	≥ 4298.3	≥ 833.75	≥ 284.08
HIP 90306	18 25 31.92	+38 21 12.67	10.5152 ± 0.0003	$\pm 1.442 \pm 0.02$	K6V	23.6 ± 0.01	Li UL	≥ 4254.11	≥ 771.96	≥ 220.21
HIP 93072	18 57 32.65	-19 02 46.86	10.3501 ± 0.0005	$\pm 1.437 \pm 0.004$	K9Vk	25.79 ± 0.01	Li UL	≥ 4285.25	≥ 815.5	≥ 266.15
HIP 98007	19 55 01.54	+04 03 43.66	9.0905 ± 0.0001	0.978 ± 0.003	K3	34.94 ± 0.21	Li UL	≥ 4510.78	≥ 1130.48	≥ 561.44
HIP 99969	20 16 55.42	+06 55 18.29	9.2486 ± 0.0001	1.14 ± 0.015	K4V	42.09 ± 0.11	Li UL	≥ 4388.19	≥ 959.42	≥ 405.74
HIP 100133	20 18 45.79	-00 39 26.33	10.2561 ± 0.0003	$\pm 1.43 \pm 0.012$	M0V	27.31 ± 0.27	Li UL	≥ 4281.86	≥ 810.76	≥ 261.29

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Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 101852	20 38 21.10	-04 30 42.73	7.6339 ± 0.0003	0.597 ± 0.012	G0V	43.59 ± 0.09	Li UL, Gyro, 2050.45 R'HK		1262.36– 2646.59	580.98–3021.11
HIP 102119	20 41 42.24	-22 19 20.51	9.3352 ± 0.0013	1.121 ± 0.009	K6Ve	24.2 ± 0.02	Li UL	≥ 4347.61	≥ 902.7	≥ 349.81
HIP 104687	21 12 22.60	-15 00 00.36	8.0059 ± 0.0007	0.654 ± 0.041	G3V	46.49 ± 0.05	Li, R'HK	5895.89	2829.05– 9452.36	855.62– 12158.07
HIP 105504	21 22 07.77	-10 30 47.89	9.7045 ± 0.0002	1.255 ± 0.015	K7	41.95 ± 0.06	Li UL	≥ 4341.38	≥ 893.98	≥ 343.5
HIP 107317	21 44 12.99	+06 38 29.28	10.9909 ± 0.0006	1.522 ± 0.021	M3.0Ve	20.57 ± 0.02	Li UL	≥ 4199.9	≥ 696.16	≥ 131.93
HIP 108036	21 53 17.77	-13 33 06.36	4.9792 ± 0.0009	0.378 ± 0.011	F2V	26.78 ± 0.08	Li, CMD	641.55	361.52–1031.51	166.74–1407.73
HIP 109807	22 14 26.74	+02 42 24.12	9.8969 ± 0.0003	1.252 ± 0.005	K5V	33.51 ± 0.02	Li UL	≥ 4335.15	≥ 885.27	≥ 334.91
HIP 110106	22 18 13.19	-03 10 19.84	10.5093 ± 0.0003	1.399 ± 0.02	K7V	36.11 ± 0.09	Li UL	≥ 4272.52	≥ 797.7	≥ 247.79
HIP 110401	22 21 43.34	-17 09 40.90	10.2378 ± 0.0005	1.208 ± 0.013	K4V	47.48 ± 0.13	Li UL	≥ 4364.59	≥ 926.43	≥ 374.95
HIP 110663	22 25 04.26	-13 59 53.92	8.4368 ± 0.0007	0.77 ± 0.005	G8/K0V	43.29 ± 0.05	Li UL, Gyro, 860.94 R'HK		529.79–2168.83	392.82–2847.0
HIP 111976	22 40 53.96	+64 54 26.04	11.2184 ± 0.0003	1.548 ± 0.1	K5	45.83 ± 0.03	Li UL	≥ 4206.41	≥ 705.19	≥ 126.14
HIP 113159	22 54 54.77	-22 22 08.23	6.6536 ± 0.0004	0.469 ± 0.005	F7V	43.5 ± 0.05	Li, R'HK, 600.81 CMD		533.5–751.18	358.69–1086.46
HIP 115280	23 20 53.26	+38 10 56.36	5.6536 ± 0.0004	0.468 ± 0.004	F5V	42.05 ± 0.06	Li UL, R'HK, 1499.09 CMD		1124.39– 1920.51	825.97–2277.7
HIP 115411	23 22 43.20	-19 41 25.93	8.5252 ± 0.0003	0.7 ± 0.005	G6V	42.88 ± 0.04	Li UL, R'HK	9476.12	6862.42– 11765.94	4396.19– 12807.11

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Target	R.A.	Dec.	G	$B - V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 116973	23 42 43.81	-19 52 57.32	9.719 ± 0.0004	1.085 ± 0.008	K3V	41.93 ± 0.06	Li UL	≥ 4468.03	≥ 1071.06	≥ 525.19
HIP 117159	23 45 09.93	+29 33 42.72	8.147 ± 0.0002	0.859 ± 0.006	K0	29.75 ± 0.13	Li UL	≥ 4766.76	≥ 1475.94	≥ 801.54
HIP 117235	23 46 17.95	+10 16 41.79	11.1245 ± 0.0002	1.376 ± 0.133	K7V	47.8 ± 0.08	Li UL	≥ 4278.54	≥ 806.1	≥ 233.62
HIP 117410	23 48 25.69	-12 59 14.85	9.3062 ± 0.0019	1.244 ± 0.014	K5Vke	28.57 ± 0.22	Li UL, Gyro	635.14	394.14–1127.47	309.22–12215.08
HIP 117795	23 53 19.84	+59 56 42.41	10.0403 ± 0.0002	1.21 ± 0.015	K8	26.72 ± 0.01	Li UL	≥ 4346.62	≥ 901.3	≥ 350.04
HIP 118086	23 57 14.38	-16 30 27.36	10.348 ± 0.0003	1.3 ± 0.015	K7V	36.7 ± 0.03	Li UL	≥ 4291.23	≥ 823.87	≥ 273.54
HIP 118310	23 59 47.78	+06 39 50.95	8.4993 ± 0.0002	1.187 ± 0.052	K4V	23.03 ± 0.01	Li UL	≥ 4399.71	≥ 975.53	≥ 416.39

2.1. Stars within the Lithium Dip

There are several stars in our sample that are near the lithium dip, which is a range of $B - V$ ($\sim 0.4 - 0.55$) where lithium depletes rapidly around the age of the Hyades. This rapid evolution means the feature is not well modeled (e.g. ?), we discuss the consistency of the ages for stars near the lithium dip below.

Many stars with $0.4 < B - V < 0.55$ have age posteriors from different tracers that are generally consistent, including the lithium posterior. For example, HIP 42343 has ages from lithium (4964^{+4688}_{-2766} Myr), R'_{HK} (4793^{+3345}_{-2103} Myr), and CMD position (2105^{+1309}_{-1071} Myr) that give a combined posterior of 2791^{+933}_{-895} Myr. Similarly, the posteriors mainly agree for HIP 17118, HIP 12837, HIP 71899, and HIP 27021.

HIP 47013 shows significant offset between posteriors. The lithium non-detection gives a 2σ lower limit of 1528 Myr, which is consistent with the R'_{HK} posterior of 4997^{+3340}_{-2175} Myr. However, the CMD posterior gives a significantly younger age of 1181^{+247}_{-203} Myr. This may be due to the small number of stars within the lithium dip in the $\sim 500 - 1000$ Myr clusters used to calibrate **BAFFLES**. Because of this limitation, we recommend not relying on ages from lithium alone for stars within the lithium dip. We see similar discrepancies between the posteriors for HIP 6943, HIP 113159, HIP 10050, HIP 71243, HIP 3293, HIP 6776, HIP 44162, HIP 2843, HIP 115280, and HIP 21066.

REFERENCES